

Disproof of conjecture 00000000165: the sum-cover number of the d -th power subgroup is not $(1 + o(1)) \log p / (\log p - \log d)$

Jackmeson1

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1 The conjecture and the reading

Statement (English, verbatim). Conjecture: Sum covers by multiplicative subgroups: for $d \mid p-1$, the minimal k for which the k -fold sumset of the subgroup H of d -th powers in $\mathbb{F}_p^*$ covers $\mathbb{F}_p$ is $(1 + o(1)) \cdot \log p / (\log p - \log d)$ (as $d \rightarrow \infty$ with $d < p^{1-\varepsilon}$).

Chinese text. It states the same, except that it calls H the subgroup of “ k -th powers” (*k ci mi*). Since k is the number of summands being minimized, and d appears in the hypothesis $d \mid p-1$ and in the formula, this is a slip for d -th powers. We use the English reading (d -th powers) and do not exploit the slip.

Objects. p is a prime and $d \mid p-1$. $H = \{x^d : x \in \mathbb{F}_p^*\}$ is the subgroup of d -th powers; it never contains 0. The k -fold sumset is $kH = \{h_1 + \dots + h_k : h_i \in H\}$, and

$$K(p, d) = \min\{k \geq 1 : kH = \mathbb{F}_p\}, \quad L(p, d) = \frac{\log p}{\log p - \log d}.$$

Reading of the asymptotic. For each fixed $\varepsilon \in (0, 1)$: for every $\eta > 0$ there is D such that $|K(p, d)/L(p, d) - 1| < \eta$ for all primes p and all $d \mid p-1$ with $d \geq D$ and $d < p^{1-\varepsilon}$. (For $\varepsilon \geq 1$ the window $d < p^{1-\varepsilon} \leq 1$ contains no divisor d , so the claim is vacuous there; we refute it for *every* $\varepsilon \in (0, 1)$, hence both the “for all ε ” and the “for some $\varepsilon \in (0, 1)$ ” readings.)

Readings refuted (all in Lean). The literal K against L , $\lceil L \rceil$, $\lfloor L \rfloor$ and $\text{round}(L)$; and the “at most k summands” variant $K_0(p, d) = \min\{k \geq 1 : k(H \cup \{0\}) = \mathbb{F}_p\}$ against L , $\lfloor L \rfloor$ and $\text{round}(L)$. **Not refuted:** K_0 against $\lceil L \rceil$ (on our family both equal 2 may well hold), and any reading that adds a lower bound on d (such as $d \geq p^\delta$) which the text does not contain.

2 The counterexample family

Lemma 1 (Dirichlet). *For every N there are a prime p and integers $d \geq \max(N, 4)$, $q \geq 0$ with $d^d < p$ and $p-1 = d(2q+1)$.*

Proof. Let $d = 2(N+2)$. Then $d+1$ is odd and coprime to d , so $\gcd(d+1, 2d) = 1$. By Dirichlet’s theorem there is a prime $p > d^d$ with $p \equiv d+1 \pmod{2d}$. As $d+1 < 2d$, $p = 2dq + d+1$ for some q , so $p-1 = d(2q+1)$. $\square$

Fix such (p, d, q) . Then $d \mid p-1$, the index of H is d , and $|H| = (p-1)/d = 2q+1$ is odd.

Lemma 2. $-1 \notin H$. Consequently $0 \notin H$ and $0 \notin H + H$.

Proof. If $u^d = -1$ for a unit u , then by Fermat $1 = u^{p-1} = (u^d)^{2q+1} = (-1)^{2q+1} = -1$, impossible since $p > 2$. Every element of H is a unit, so $0 \notin H$. If $u^d + v^d = 0$, then $(vu^{-1})^d = -1$, a contradiction. $\square$

Lemma 3 (non-degeneracy and existence of a cover). 1 and 2^d are distinct elements of H , and $pH = \mathbb{F}_p$. Hence $K(p, d)$ and $K_0(p, d)$ are genuine minima.

Proof. $1 = 1^d$ and $2^d = (2)^d$ with 2 a unit ($p > 2$). Since $1 < 2^d \leq d^d < p$, $2^d \not\equiv 1 \pmod{p}$. So $|H| \geq 2$. The Cauchy–Davenport theorem $|A + B| \geq \min\{p, |A| + |B| - 1\}$ in $\mathbb{Z}/p\mathbb{Z}$ gives by induction $|kH| \geq \min\{p, k\}$ for $k \geq 1$, so $|pH| = p$ and $pH = \mathbb{F}_p$. Since $H \subseteq H \cup \{0\}$, also $p(H \cup \{0\}) = \mathbb{F}_p$. $\square$

Theorem 4. $K(p, d) \geq 3$ and $K_0(p, d) \geq 2$.

Proof. $1H = H$ and $2H = H + H$ miss 0 (Lemma 2), and a cover exists (Lemma 3). $1(H \cup \{0\}) = H \cup \{0\}$ misses -1 . $\square$

Lemma 5. If $d \geq 2$ and $d^4 < p$, then $\log p - \log d > 0$ and $1 < L(p, d) \leq 4/3$; hence $\lceil L \rceil = 2$ and $\lfloor L \rfloor = \text{round}(L) = 1$. If moreover $d^d < p$ and $(1 - \varepsilon)d \geq 1$, then $d < p^{1-\varepsilon}$.

Proof. $4 \log d < \log p$ gives $L \leq 4/3$; $\log d > 0$ gives $L > 1$. From $d \log d < \log p$: $(1 - \varepsilon) \log p > (1 - \varepsilon)d \log d \geq \log d$. $\square$

Theorem 6. For every $\varepsilon \in (0, 1)$ and every N there is an admissible pair (p, d) with $d \geq N$, $d \mid p - 1$, $d < p^{1-\varepsilon}$, on which $K/L \geq 9/4$, $K/\lceil L \rceil \geq 3/2$, $K/\lfloor L \rfloor = K/\text{round}(L) \geq 3$, $K_0/L \geq 3/2$, $K_0/\lfloor L \rfloor = K_0/\text{round}(L) \geq 2$. So, with $\eta = 1/2$, each asymptotic in Section 1 fails. The conjecture is false.

Proof. Apply Lemma 1 with $\max(N, \lceil 1/(1 - \varepsilon) \rceil)$; then $d \geq 4$ gives $d^4 \leq d^d < p$, and Lemma 5 and Theorem 4 give the ratios. $\square$

The family is non-degenerate: $d \rightarrow \infty$; H contains the distinct elements 1 and 2^d (proved in Lean); mathematically H has index d and $|H| = (p - 1)/d = 2q + 1 \geq 3$ (not needed for the proof); and $\log d / \log p < 1/d \rightarrow 0$, so it lies inside the window for every $\varepsilon \in (0, 1)$. Numerically, $(d, p) = (4, 269)$ and $(6, 46663)$ give $K = 3$ with $L \approx 1.33$ and 1.20 .

3 Lean formalization

File `lean/Conjecture165/Basic.lean` (namespace `C165`, import `Mathlib` only).

- `powSubgroup p d := (powMonoidHom d).range`, the subgroup of d -th powers of $(\mathbb{Z}/p)^\times$; `H p d` is its image in `ZMod p`; `mem_H`.
- The k -fold sumset is `Mathlib`'s pointwise $k \bullet S$ (`k • S`); `mem_kfold_sumset` proves $x \in k \bullet S$ iff x is a sum of k elements of S .
- `coverNumber p d := sInf {k | 1 <= k and k nsmul H p d = univ}`; `coverNumberAtMost` uses `H p d ∪ {0}`; `L p d` is the formula. (`sInf` of an empty set is `0`; on our family a cover is proved to exist, so the value is the true minimum.)

- `Asymp eps K F`: the uniform $(1 + o(1))$ statement of Section 1; `Conjecture165 := forall eps, 0 < eps -> eps < 1 -> Asymp eps coverNumber L`.
- `exists_pair` (Lemma 1, via Mathlib's `Nat.forall_exists_prime_gt_and_eq_mod`); `neg_one_not_pow`, `zero_not_mem_H`, `zero_not_mem_HH`, `neg_one_not_mem_H0`; `nsmul_eq_univ` (via `ZMod.cauchy_davenport`), `one_two_pow_mem_H`, `cover_exists`; `three_le_coverNumber`, `two_le_coverNumberAtMost`; `L_bounds`, `window`, `floor_round_L`.
- `family`: Theorem 6's family with all its properties.
- `main eps`: for $0 < \varepsilon < 1$, the seven negations `not Asymp eps coverNumber F` for $F \in \{L, \lceil L \rceil, \lfloor L \rfloor, \text{round } L\}$ and `not Asymp eps coverNumberAtMost F` for $F \in \{L, \lfloor L \rfloor, \text{round } L\}$.
- `not_conjecture165 : not Conjecture165`.

Axiom audit. #print axioms for `C165.not_conjecture165`, `C165.main` and `C165.family` reports only `propext`, `Classical.choice`, `Quot.sound`. No sorry, no `native_decide`, no added axioms.

4 Sources

- Dirichlet's theorem (used through Mathlib's `Nat.forall_exists_prime_gt_and_eq_mod`): Wikipedia, "Dirichlet's theorem on arithmetic progressions": "there are infinitely many primes that are congruent to a modulo d". https://en.wikipedia.org/wiki/Dirichlet%27s_theorem_on_arithmetic_progressions
- Cauchy–Davenport (used through Mathlib's `ZMod.cauchy_davenport`): Wikipedia, "Restricted sumset": it "asserts that for any prime p and nonempty subsets A and B of the prime-order cyclic group $\mathbb{Z}/p\mathbb{Z}$, $|A + B| \geq \min\{p, |A| + |B| - 1\}$ ". https://en.wikipedia.org/wiki/Restricted_sumset